



线性代数第=次作业

7. (1) 解:

$$D_n = \begin{vmatrix} x & y & 0 & \cdots & 0 & 0 \\ 0 & x & y & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & x & y \\ y & 0 & 0 & \cdots & 0 & x \end{vmatrix} = \begin{vmatrix} x+y & y & 0 & \cdots & 0 & 0 \\ x+y & x & y & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ x+y & 0 & 0 & \cdots & x & y \\ x+y & 0 & 0 & \cdots & 0 & x \end{vmatrix}$$

$$= (x+y) \begin{vmatrix} 1 & y & 0 & \cdots & 0 & 0 \\ 1 & x & y & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 1 & 0 & 0 & \cdots & x & y \\ 1 & 0 & 0 & \cdots & 0 & x \end{vmatrix} = (x+y) \begin{vmatrix} 1 & y & 0 & \cdots & 0 & 0 \\ 0 & x-y & y & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & -y & 0 & \cdots & x & y \\ 0 & -y & 0 & \cdots & 0 & x \end{vmatrix}$$

$$= (x+y) \begin{vmatrix} x-y & y & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ -y & 0 & \cdots & x & y \\ -y & 0 & \cdots & 0 & x \end{vmatrix} = (x+y) \begin{vmatrix} x & y & \cdots & 0 & 0 \\ x & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ x & 0 & \cdots & x & y \\ x-y & 0 & \cdots & 0 & x \end{vmatrix}$$

上述过程省略。

$$\text{按第一列展开: } D_n = x \begin{vmatrix} x & y & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & x & y \\ 0 & 0 & \cdots & 0 & x \end{vmatrix} + (-1)^{n+1} y \begin{vmatrix} y & 0 & \cdots & 0 & 0 \\ x & y & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & x & y \end{vmatrix}$$

$$\text{记为 } D_n = x A_n + (-1)^{n+1} y B_n$$

对 A_n 总按第一列展开, B_n 总按第一行展开, 则有:

$$D_n = x^n + (-1)^{n+1} y \cdot y^{n-1} = x^n + (-1)^{n+1} y^n$$

$$\text{(2) 解: } D_n \xrightarrow[r_2-r_1]{r_i-r_1, 2 \leq i \leq n} \begin{vmatrix} 1 & 1 & \cdots & 1 \\ 0 & -1 & 0 & \cdots & 0 \\ 0 & 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & -1 \end{vmatrix} = \begin{vmatrix} -1 & 0 & \cdots & 0 \\ 0 & -1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & -1 \end{vmatrix} = (-1)^{n-1}$$

(3) 解:

$$D_n \xrightarrow[r_2-r_1]{r_i-r_1, 1 \leq i \leq n-1} \begin{vmatrix} x-a & 0 & 0 & \cdots & a-x \\ 0 & x-a & 0 & \cdots & a-x \\ 0 & 0 & x-a & \cdots & a-x \\ \vdots & \vdots & \vdots & & \vdots \\ a & a & a & \cdots & x \end{vmatrix} = \begin{vmatrix} xa & 0 & 0 & \cdots & 0 \\ 0 & x-a & 0 & \cdots & 0 \\ 0 & 0 & x-a & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ a & a & a & \cdots & x+ac_{n-1} \end{vmatrix}$$

$$= (x+ac_{n-1}) (-1)^{n-1} (x-a)^{n-1}$$

$$= (x+ac_{n-1}) (x-a)^{n-1}$$

$$\begin{aligned}
 (4) \quad D_n &= \begin{vmatrix} -1 & 0 & 0 & \cdots & 0 \\ 2 & 2 & 2 & \cdots & 2 \\ 0 & 0 & 1 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & n-2 \end{vmatrix} = (-1) \begin{vmatrix} 1 & 2 & 2 & \cdots & 2 \\ 0 & 0 & 1 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & n-2 \end{vmatrix} \\
 &= (-2) \begin{vmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 2 & 0 & \cdots & 0 \\ 0 & 0 & 3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & n-2 \end{vmatrix} = (-2) \times 2 \times 3 \times \cdots (n-2) = (-2)(n-2)!
 \end{aligned}$$

5

$$\begin{aligned}
 (5) \quad D_n &= 2 \times \begin{vmatrix} 2 & 1 & & 0 \\ 1 & 2 & 1 & \\ & 1 & 2 & \\ & & & 2 & 1 \\ 0 & & & 1 & 2 \end{vmatrix} + (-1) \begin{vmatrix} 1 & 1 & & 0 \\ 0 & 2 & 1 & \\ & 1 & 2 & \\ & & & 2 & 1 \\ 0 & & & 1 & 2 \end{vmatrix} \\
 &= 2D_{n-1} - D_{n-2}, \text{ 数列特征方程为 } x^2 - 2x - 1 = 0 \Rightarrow \text{有唯一解 } x=1 \\
 \therefore D_2 &= \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 3 \quad \therefore D_n = n+1
 \end{aligned}$$

10

$$(6) \quad D_n = \begin{vmatrix} -a_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & -a_2 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -a_{n-1} & 0 \\ 1 & 2 & 3 & \cdots & n-1 & n \end{vmatrix} = (-1)^{n-1} \begin{vmatrix} -a_1 & & & & \\ & -a_2 & & & \\ & & -a_3 & & \\ & & & \ddots & \\ & & & & -a_{n-1} \end{vmatrix}$$

15

$$= (-1)^{n-1} a_1 a_2 \cdots a_{n-1}$$

$$\begin{aligned}
 8 \text{ 的解: } D &= \begin{vmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 2 & 0 & -1 \\ 2 & 0 & 1 & -3 \end{vmatrix} = 4 \quad D_2 = \begin{vmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & -1 \\ 2 & 1 & 1 & -3 \end{vmatrix} = -4 \\
 D_1 &= \begin{vmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ -1 & 2 & 0 & -1 \\ 1 & 0 & 1 & -3 \end{vmatrix} = 8 \quad D_3 = \begin{vmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 2 & -1 & -1 \\ 2 & 0 & 1 & -3 \end{vmatrix} = 0 \\
 D_4 &= \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & -1 \\ 2 & 0 & 1 & 1 \end{vmatrix} = 4
 \end{aligned}$$

20

$$\therefore \text{存在唯一一组解: } x_1=2, x_2=-1, x_3=0, x_4=1$$

25

$$12) \quad D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & w & w^2 \\ 1 & w^2 & w \end{vmatrix} = w^2 + w^2 + w^2 - w - w - w^4 = 3w^2 - 3w$$

$$D_1 = \begin{vmatrix} 1 & 1 & 1 \\ w & w & w^2 \\ w^2 & w^2 & w \end{vmatrix} = 0, \quad D_2 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & w & w^2 \\ 1 & w^2 & w \end{vmatrix} = 3w^2 - 3w$$

$$D_3 = \begin{vmatrix} 1 & 1 & 1 \\ w & w & w^2 \\ w & w^2 & w^2 \end{vmatrix} = 0$$

$\therefore$ 存在唯一一组解: $x_1 = 0, x_2 = 1, x_3 = 0$

$$9. \text{ 解: } D = \begin{vmatrix} \lambda & 3 & 4 \\ -1 & \lambda & 0 \\ 0 & \lambda & 1 \end{vmatrix} = \lambda^3 - 4\lambda - (-3) = \lambda^3 - 4\lambda + 3 \neq 0$$

$\therefore \lambda \neq 1$ 且 $\lambda \neq 3$ 时仅有零解

$$10. \text{ 解: } D = \begin{vmatrix} \lambda & 1 & 1 \\ 1 & \mu & 1 \\ 1 & 2\mu & 1 \end{vmatrix} = \lambda\mu + 2\mu - \mu - 2\mu\lambda$$

$$= -\lambda\mu + \mu = \mu(1-\lambda) = 0$$

$\therefore \mu = 0$ 或 $\lambda = 1$ 或两者同时成立时有非零解

$$11. \text{ 解: } D = \begin{vmatrix} 1 & 1 & 1 & a \\ 1 & 2 & 1 & 1 \\ 1 & 1 & -3 & 1 \\ 1 & 1 & a & b \end{vmatrix} = \begin{vmatrix} 0 & -1 & 0 & a-1 \\ 1 & 2 & 1 & 1 \\ 0 & -1 & -4 & 0 \\ 0 & -1 & a-1 & b-1 \end{vmatrix}$$

$$= (-1)^3 \times 1 \times \begin{vmatrix} -1 & 0 & a-1 \\ -1 & -4 & 0 \\ -1 & a-1 & b-1 \end{vmatrix} = (-1) \times (4(b-1) - (a-1)^2 - 4(a-1))$$

$$= -(4b - 4 - a^2 + 2a - 1 - 4a + 4) = -4b + a^2 + 2a + 1 = 0$$

$\therefore (a+1)^2 = 4b$ 时有非零解。

Ch1-2 作业：习题一：7,8,9,10,11

*7 有少量同学算错，8 有不少同学算错，或者中间结果错但最终结果正确，如 8(2) $\Delta_1=\Delta_3=0$ ， $\Delta_2=\Delta=3\omega^2-2\omega$ 。

7(1)有些同学将该题做复杂了，可如下：

$$\text{解： } D_n = \begin{vmatrix} x & y & & \\ & x & \ddots & \\ & & \ddots & y \\ y & & & x \end{vmatrix} \stackrel{c_1 \text{ 展开}}{=} x \begin{vmatrix} x & y & & \\ & x & \ddots & \\ & & \ddots & y \\ & & & x \end{vmatrix}_{n-1} + (-1)^{n+1} y \begin{vmatrix} y & & & \\ x & y & & \\ & \ddots & \ddots & \\ & & x & y \end{vmatrix}_{n-1} = x^{n-1} + (-1)^{n+1} y^{n-1}.$$

7(3)有多种解法，见如下：

$$\text{解： } D_n = \begin{vmatrix} x & a & \cdots & a \\ a & x & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ a & a & \cdots & x \end{vmatrix} \stackrel{c_1+c_2+\cdots+c_n}{=} \begin{vmatrix} x+(n-1)a & a & \cdots & a \\ x+(n-1)a & x & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ x+(n-1)a & a & \cdots & x \end{vmatrix} \stackrel{r_1-r_1}{=} \begin{vmatrix} x+(n-1)a & a & \cdots & a \\ 0 & x-a & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & x-a \end{vmatrix} = (x+(n-1)a)(x-a)^{n-1}.$$

$$\text{解法二： } D_n = \begin{vmatrix} x & a & \cdots & a \\ a & x & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ a & a & \cdots & x \end{vmatrix} \stackrel{c_1+c_2+\cdots+c_n}{=} \begin{vmatrix} x+(n-1)a & a & \cdots & a \\ x+(n-1)a & x & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ x+(n-1)a & a & \cdots & x \end{vmatrix} \stackrel{r_1-r_1}{=} \begin{vmatrix} 1 & a & \cdots & a \\ 1 & x & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a & \cdots & x \end{vmatrix} \stackrel{c_1-a c_1}{=} \begin{vmatrix} 1 & 0 & \cdots & 0 \\ 1 & x-a & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \cdots & x-a \end{vmatrix} = (x+(n-1)a)(x-a)^{n-1}.$$

$$\text{解法三： } D_n = \begin{vmatrix} x & a & \cdots & a \\ a & x & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ a & a & \cdots & x \end{vmatrix} \stackrel{c_1-c_1}{=} \begin{vmatrix} x & a-x & \cdots & a-x \\ a & x-a & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a & 0 & \cdots & x-a \end{vmatrix} \stackrel{r_1+r_2+\cdots+r_n}{=} \begin{vmatrix} x+(n-1)a & 0 & \cdots & 0 \\ a & x-a & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a & 0 & \cdots & x-a \end{vmatrix} = (x+(n-1)a)(x-a)^{n-1}.$$

7(5) 很多同学都先求出递推式 $D_n=2D_{n-1}-D_{n-2}$ ，再用等差数列 $D_n-D_{n-1}=D_{n-1}-D_{n-2}=\cdots=D_2-D_1=1, D_1=2$ 计算，也可如下：

$$\text{解： } D_n = \begin{vmatrix} 2 & 1 & & \\ 1 & 2 & \ddots & \\ & \ddots & \ddots & 1 \\ & & & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 & & \\ 1 & 2 & \ddots & \\ & \ddots & \ddots & 1 \\ & & & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 & & \\ 0 & 2 & \ddots & \\ & \ddots & \ddots & 1 \\ & & & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & & \\ 1 & 1 & \ddots & \\ & \ddots & \ddots & 0 \\ & & & 1 \end{vmatrix} + D_{n-1} = 1 + D_{n-1} = \cdots = n-1 + D_1 = n+1.$$

$$\text{解法二： } D_n = \begin{vmatrix} 2 & 1 & & \\ 1 & 2 & \ddots & \\ & \ddots & \ddots & 1 \\ & & & 1 \end{vmatrix} \stackrel{r_{i+1}-\frac{i}{i+1}r_i}{=} \begin{vmatrix} 2 & 1 & & \\ 0 & 3/2 & \ddots & \\ & \ddots & \ddots & 1 \\ & & & 0 \end{vmatrix} \stackrel{i=1,2,\dots,n-1}{=} 2 \times \frac{3}{2} \times \frac{4}{3} \times \cdots \times \frac{n+1}{n} = n+1.$$

7(6)有多种解法，见如下：

$$\text{解： } D_n = \begin{vmatrix} -a_1 & a_1 & 0 & \cdots & 0 & 0 \\ 0 & -a_2 & a_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -a_{n-1} & a_{n-1} \\ 1 & 1 & 1 & \cdots & 1 & 1 \end{vmatrix} \stackrel{c_1+c_2+\cdots+c_n}{=} \begin{vmatrix} 0 & a_1 & 0 & \cdots & 0 & 0 \\ 0 & -a_2 & a_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -a_{n-1} & a_{n-1} \\ n & 1 & 1 & \cdots & 1 & 1 \end{vmatrix} \stackrel{c_1 \text{ 展开}}{=} (-1)^{n+1} n \begin{vmatrix} a_1 & & & & \\ -a_2 & a_2 & & & \\ & \ddots & \ddots & & \\ 0 & 0 & a_{n-1} & a_{n-1} \end{vmatrix} = (-1)^{n+1} n \prod_{i=1}^{n-1} a_i.$$

$$\text{解法二： } D_n = \begin{vmatrix} -a_1 & a_1 & 0 & \cdots & 0 & 0 \\ 0 & -a_2 & a_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -a_{n-1} & a_{n-1} \\ 1 & 1 & 1 & \cdots & 1 & 1 \end{vmatrix} \stackrel{c_n+c_1+\cdots+c_{n-1}}{=} \begin{vmatrix} -a_1 & a_1 & 0 & \cdots & 0 & 0 \\ 0 & -a_2 & a_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -a_{n-1} & 0 \\ 1 & 1 & 1 & \cdots & 1 & n \end{vmatrix} \stackrel{c_n \text{ 展开}}{=} n \begin{vmatrix} -a_1 & a_1 & & & \\ & -a_2 & a_2 & & \\ & & \ddots & \ddots & \\ & & & -a_{n-1} & \\ & & & & -a_{n-1} \end{vmatrix} = (-1)^{n-1} n \prod_{i=1}^{n-1} a_i.$$

$$\text{解法三： } D_n = \begin{vmatrix} -a_1 & a_1 & 0 & \cdots & 0 & 0 \\ 0 & -a_2 & a_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -a_{n-1} & a_{n-1} \\ 1 & 1 & 1 & \cdots & 1 & 1 \end{vmatrix} \stackrel{c_{i+1}+c_i}{=} \begin{vmatrix} -a_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & -a_2 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -a_{n-1} & 0 \\ 1 & 2 & 3 & \cdots & n-1 & n \end{vmatrix} = (-1)^{n-1} n \prod_{i=1}^{n-1} a_i.$$

其余一些作业解题思路：

7(2) $r_i-r_1, i=2,3,\dots,n$; (4) $r_i-r_2, i=3,4,\dots,n$ ，然后 c_1-c_2 或者块三角降阶

8(1) $\Delta=4, \Delta_1=8, \Delta_2=-4, \Delta_3=0, \Delta_4=4$; (2) $\Delta=-3\sqrt{3}i$ 或 $\Delta=3(\omega^2-\omega) \neq 0, \Delta_1=0, \Delta_2=\Delta, \Delta_3=0$

9 $\Delta=(\lambda-1)(\lambda-3) \neq 0$

10 $\Delta=-\mu(\lambda-1)=0$

11 $\Delta=(a+1)^2-4b=0$